FINAL PROJECT REPORT Internship Program: Data Science & Machine Learning Organization: Upskill Campus

1. Student Details Name: Shaik Mohammed Amaan Project Title: Fake News Detection using Big Data Analytics Domain: Machine Learning / Big Data Submission Date: [13/04/2026]
2. Project Abstract The rise of social media has led to an explosion of misinformation. This project aims to build a robust classification system that can automatically detect whether a news article is "Real" or "Fake." By utilizing Natural Language Processing (NLP) and the Passive Aggressive Classifier, the model achieves high accuracy by learning from massive datasets of labeled news content.
3. System Architecture The project follows a standard Data Science pipeline to ensure data integrity and model performance. Data Acquisition: Using a dataset containing thousands of news articles (Text, Title, Label). Preprocessing: Tokenization and removal of stop words to clean the data. Vectorization: Converting text into numerical form using TF-IDF (Term Frequency-Inverse Document Frequency). Classification: Training the model using the Passive Aggressive Classifier algorithm.
4. Technical Implementation 4.1 Tools & Technologies Language: Python 3.x Environment: Google Colab / VS Code Key Libraries: Pandas (Data manipulation), Scikit-learn (Machine Learning), NumPy (Numerical operations). 4.2 Mathematical Basis The TF-IDF score for a term t in document d is calculated as:tf-idf(t,d)=tf(t,d)*idf(t)Where idf(t) = \log\frac{N}{df(t)}, with N being the total number of documents.
5. Results and Observations Model Performance: The system achieved an accuracy of approximately 94%. Confusion Matrix: The model shows a low false-positive rate, meaning real news is rarely flagged as fake. Challenges: Handling highly biased articles that use factual data but misleading context was the primary challenge.
6. Conclusion This internship project successfully demonstrates how Big Data Analytics and Machine Learning can be leveraged to maintain information integrity. The model is lightweight, fast, and capable of being deployed as a real-time web application or browser extension for automated fact-checking.